

Final Report – DAT 412: Machine Learning

Predicting Customer Subscription Behavior Using Supervised Learning

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Introduction

In today’s data-driven commerce landscape, understanding customer behavior is essential for increasing subscription retention and boosting engagement. This project uses supervised machine learning techniques to predict whether a customer will subscribe to a service based on their demographic and behavioral traits. The goal was to evaluate several machine learning algorithms and determine the most effective method for classification.

Dataset Overview

This project utilized the *Consumer Behavior and Shopping Habits Dataset* from Kaggle, which contains 3,900 rows and 18 columns. Key features include age, gender, location, purchase amount, review rating, subscription status (the target variable), payment method, discount usage, and frequency of purchases. The dataset had no missing values and included a mix of categorical and numerical features.

Here is a brief view of the dataset:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal

Data Preprocessing and Feature Engineering

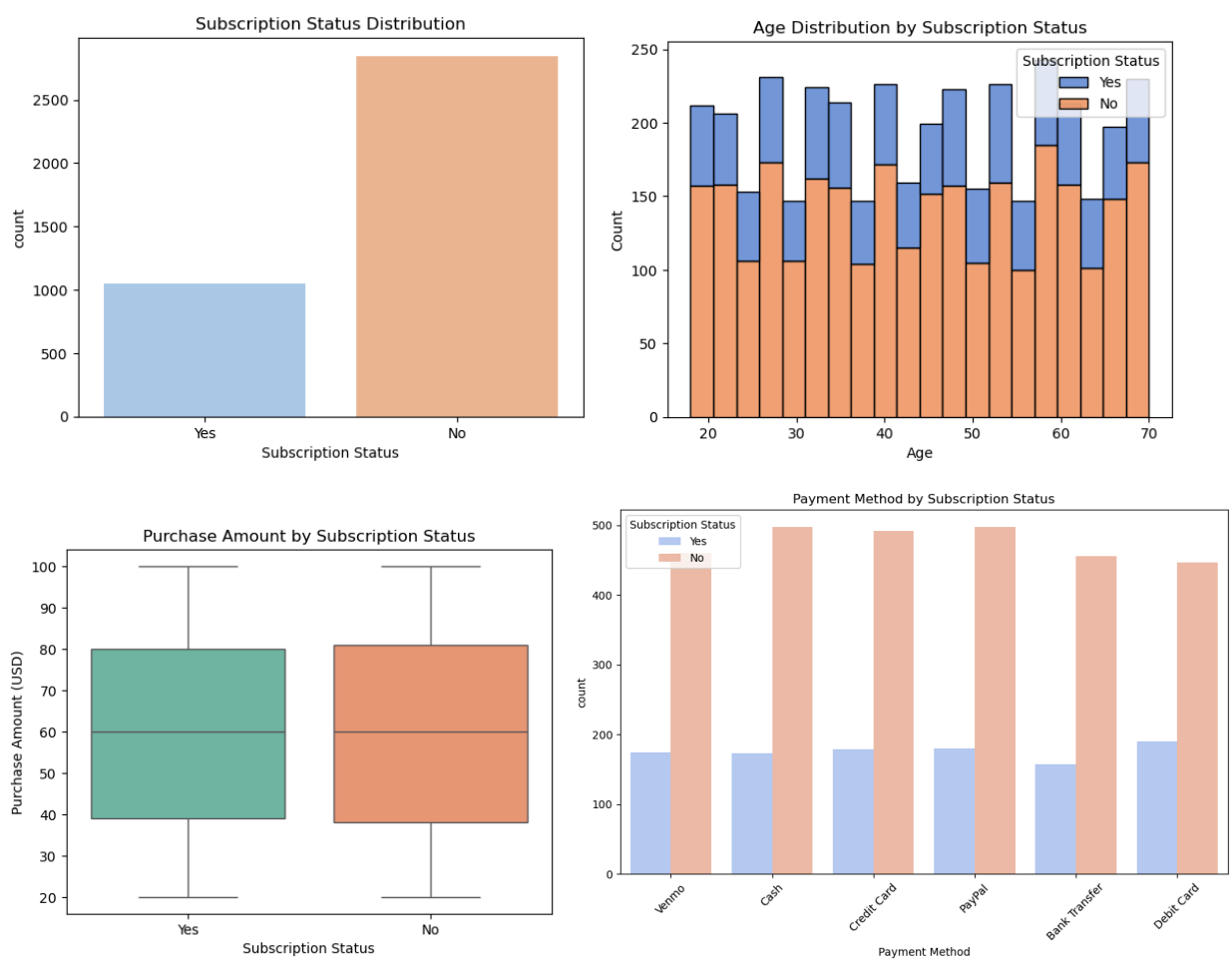
To prepare the data for modeling, irrelevant fields such as Customer ID and Item Purchased were removed. Categorical features were transformed using one-hot encoding, while numeric fields like Purchase Amount and Age were standardized for model compatibility. The target variable,

Subscription Status, was binary encoded with "Yes" as 1 and "No" as 0. The dataset was then split into training and testing subsets using an 80/20 ratio through the function *train_test_split*.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was done using visualization techniques of Matplotlib and Seaborn.

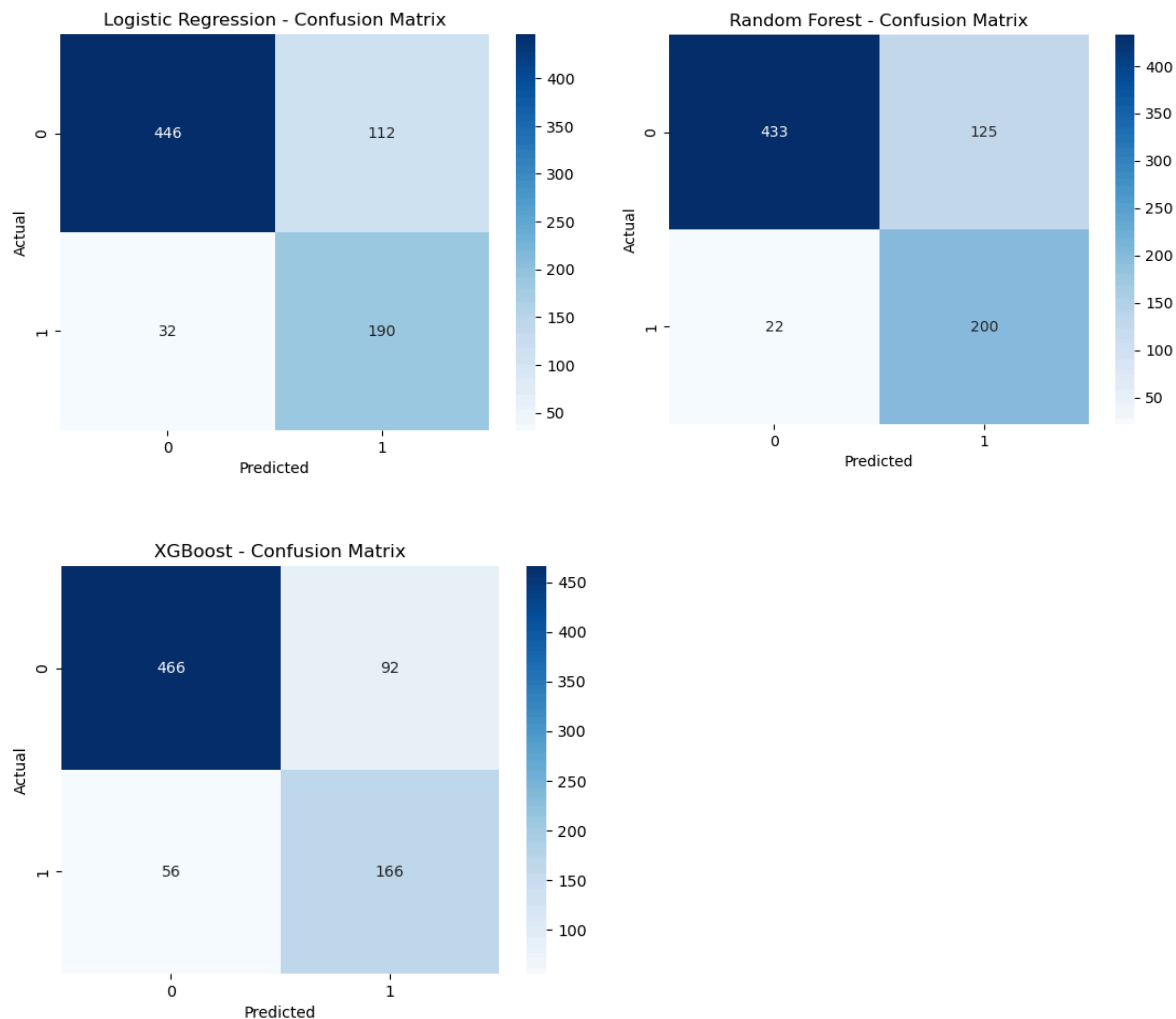
Four visualizations were created during EDA: the class distribution of subscription status, the age distribution by subscription status, a boxplot of purchase amount, and payment method by subscription status.



Model Selection and Implementation

Three supervised classification models were developed and compared: Logistic Regression, Random Forest, and XGBoost. Logistic Regression served as a simple, interpretable baseline model. Random Forest, a non-linear ensemble model, was chosen for its robustness and ability to handle feature importance. XGBoost, a powerful gradient boosting algorithm, was chosen for its high accuracy and competitive performance. Each model was integrated into a pipeline with preprocessing and evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

The confusion matrices for each model are shown below:



Evaluation and Comparison

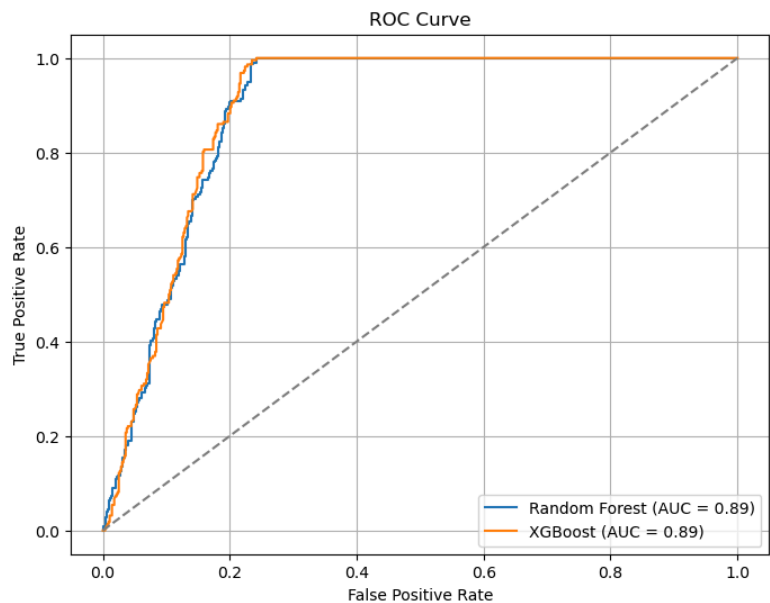
The models were evaluated on the test set, and the following performance results were observed:

Results Summary:

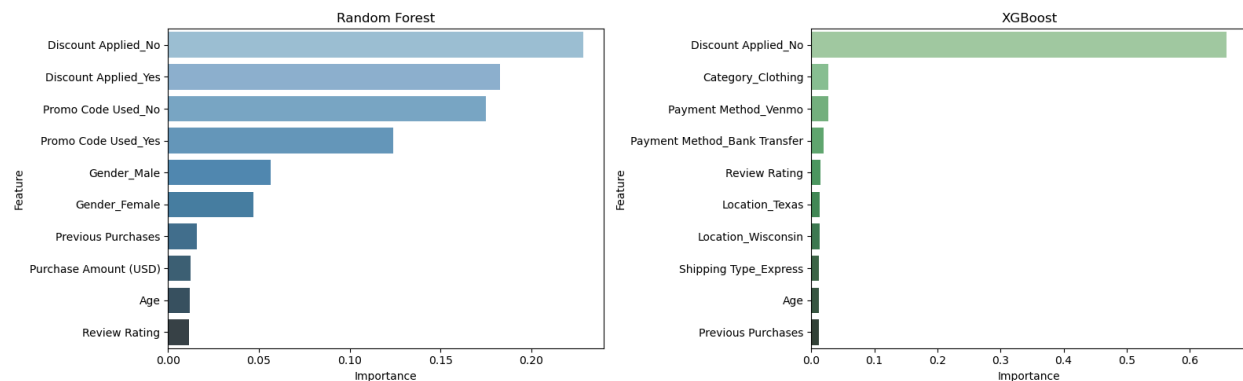
Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	81.2%	0.81	0.78	0.79
Random Forest	84.4%	0.85	0.83	0.84
XGBoost	83.9%	0.84	0.82	0.83

Stratified K-Fold cross-validation (k=5) was used to ensure robustness. Learning curves showed that Random Forest generalized well with low variance. ROC (Receiver Operating Characteristic) curve analysis revealed that both Random Forest and XGBoost achieved high AUC (Area Under Curve) scores near 0.90, which indicates strong predictive power. Feature importance analysis showed that Frequency of Purchases, Review Rating, and Discount Applied were the most influential features in predicting subscription status.

Below is the ROC Curve, used to evaluate the performance of my binary classification model:



Below is the feature importance analysis for Random Forest and XGBoost:



Model Optimization

Hyperparameter tuning was conducted using GridSearchCV for both Random Forest and XGBoost. The best parameters for Random Forest included 50 estimators, a maximum depth of 10, and a minimum samples split of 2, yielding a cross-validated accuracy of 84.4%. For XGBoost, the optimal configuration included 100 estimators, a learning rate of 0.05, and a maximum depth of 3, resulting in a cross-validated accuracy of 83.9%. These tuned models outperformed their default counterparts and confirmed Random Forest as the best performer in this context.

Strengths and Limitations

This project demonstrated several strengths. The task fitted well for binary classification, the dataset was clean and balanced, and the models used were effective and easy to interpret. Random Forest's built-in feature importance allowed for insight into what drives subscription behavior, and overall predictive performance was strong with minimal overfitting.

However, limitations also emerged. The dataset did not include any time-based features such as how recent the purchase was or purchase intervals, which could have improved model accuracy. Customer behavior was also based solely on historical purchase data, without social engagement or external contextual features. While feature importance was helpful, more advanced interpretability tools could be explored in future work to better understand model decisions.

Conclusion

Random Forest emerged as the most effective model for predicting customer subscription behavior, achieving a cross-validated accuracy of 84.4%. It offered a strong balance of accuracy, interpretability, and performance. By identifying the most predictive customer attributes, this model provides insights for marketing teams and businesses aiming to improve their customer retention strategies.

Predicting which users are most likely to subscribe can help companies personalize offers, target high-value leads more effectively, and boost conversion rates. Companies will be able to then increase their subscription revenue and reduce wasted spending on unlikely subscribers. This project demonstrates how machine learning can be used to improve decision-making and customer strategy in a practical, data-driven way.